

WETASKIWIN AGCM, AB, Canada**WMO: 718950****Lat: 52.9803N****Lon: 113.4442W****Elev: 774****StdP: 92.37****Time Zone: -7.00 (NAM)****Period: 08-19****WBAN: 99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-30.8	-27.8	-33.9	0.2	-30.6	-31.0	0.2	-27.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	27.7	17.1	26.2	16.4	24.5	15.7	19.2	25.1	18.0	24.1	16.9	22.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	13.3	22.7	15.7	12.2	21.0	14.6	11.4	19.8	58.2	25.3	53.9	24.2	50.4	22.6	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	DB	-35.7	31.1	3.1	1.9	-38.0	32.5	-39.8	33.6	-41.6	34.7	-43.8	36.0	(4)
(5)			WB	-35.8	21.5	3.1	1.4	-38.1	22.5	-39.9	23.3	-41.7	24.0	-43.9	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-9.5	-10.8	-4.7	3.0	10.3	14.1	16.2	15.3	10.6	3.8	-4.5	-11.3	(6)	
	DBStd	11.69	8.24	8.33	7.71	5.24	4.37	2.88	2.70	3.01	4.40	4.63	7.06	7.74	(7)	
	HDD10.0	3249	604	582	456	216	49	4	0	2	45	199	435	659	(8)	
	HDD18.3	5694	862	816	714	462	248	128	77	99	234	452	685	917	(9)	
	CDD10.0	619	0	0	0	4	60	128	192	167	62	6	1	0	(10)	
	CDD18.3	21	0	0	0	0	1	3	9	7	1	0	0	0	(11)	
	CDH23.3	608	0	0	0	2	62	73	196	180	93	3	0	0	(12)	
	CDH26.7	99	0	0	0	0	8	8	33	31	20	0	0	0	(13)	
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)	
(15) Precipitation	PrecAvg	471	22	15	21	28	53	88	90	59	35	22	21	17	(15)	
	PrecMax	575	73	37	46	67	104	145	178	112	76	55	60	34	(16)	
	PrecMin	341	2	3	5	6	19	26	26	17	2	5	4	4	(17)	
	PrecStd	63	14	8	10	14	22	28	36	20	20	13	13	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	7.0	12.8	22.4	27.7	27.8	29.7	29.6	29.4	22.5	15.0	6.1	(19)	
		MCWB	3.0	3.2	6.1	10.4	13.0	17.1	18.5	17.9	15.7	11.7	7.9	2.5	(20)	
	2%	DB	5.6	4.4	9.2	18.6	25.0	25.4	27.4	27.1	26.2	17.8	9.7	3.9	(21)	
		MCWB	2.3	1.3	4.2	8.7	11.9	15.5	18.7	17.6	14.5	9.7	5.0	1.1	(22)	
	5%	DB	4.0	2.7	6.6	14.9	22.6	23.5	25.6	25.3	23.2	14.6	6.3	1.8	(23)	
		MCWB	1.5	0.3	2.9	6.6	11.3	14.5	17.9	17.2	13.7	7.7	2.8	-0.6	(24)	
	10%	DB	2.0	0.9	4.3	12.1	19.9	21.7	23.7	23.4	19.9	11.9	4.2	-0.4	(25)	
		MCWB	-0.1	-1.0	1.3	5.4	10.2	13.7	16.8	16.5	12.4	6.5	1.6	-2.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.2	6.6	10.8	14.2	18.7	21.1	20.4	16.6	12.3	8.0	2.9	(27)	
		MCDB	6.4	6.5	11.5	21.6	23.9	25.3	27.1	26.2	25.6	21.0	14.3	5.7	(28)	
	2%	WB	2.5	1.7	4.6	8.7	12.9	16.8	19.6	19.1	15.4	10.2	5.3	1.2	(29)	
		MCDB	5.2	4.1	8.7	17.9	22.0	23.1	25.3	25.1	24.3	16.7	8.9	3.6	(30)	
	5%	WB	1.5	0.4	3.0	7.2	12.0	15.6	18.4	17.9	14.1	8.4	3.2	-0.4	(31)	
		MCDB	3.9	2.4	6.3	13.9	21.0	21.2	24.1	24.1	22.0	13.3	5.8	1.7	(32)	
	10%	WB	0.0	-0.9	1.5	5.7	10.9	14.5	17.3	16.7	12.9	7.0	1.8	-2.2	(33)	
		MCDB	1.8	1.0	3.9	11.3	18.9	20.0	22.8	22.4	19.2	11.1	3.8	-0.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	11.0	10.6	11.8	13.5	12.1	12.7	13.6	14.0	12.1	9.9	9.6	(35)	
		MCDBR	11.3	11.4	13.1	17.8	18.0	15.8	16.3	17.7	20.7	17.5	12.9	10.9	(36)	
	5% WB	MCWBR	8.8	8.8	8.5	9.2	7.7	7.4	8.7	9.2	9.9	9.7	8.7	8.7	(37)	
		MCWBR	11.1	11.1	12.6	16.8	15.7	13.3	15.1	15.9	18.5	16.0	11.7	10.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.259	0.264	0.302	0.324	0.340	0.347	0.350	0.360	0.308	0.282	0.253	0.243	(40)	
		taud	2.308	2.383	2.372	2.388	2.394	2.391	2.421	2.354	2.487	2.538	2.462	2.312	(41)	
	Ebn at Noon		740	864	891	912	910	903	893	862	878	838	763	694	(42)	
		Edh at Noon	59	74	94	105	110	112	107	108	83	64	50	48	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.03	2.08	3.56	4.70	5.75	6.06	6.27	5.14	3.56	2.09	1.03	0.76	(44)	
	RadStd		0.10	0.17	0.27	0.45	0.39	0.39	0.32	0.35	0.38	0.24	0.11	0.07	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page